

A simulation loop for ecological study design and inference: from mechanistic hypotheses to precision-based conclusions

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Abstract

A core skill in empirical science is to separate pattern from noise: judging whether an apparent signal in the data reflects a reproducible process or the combined effects of variability, measurement limitations, sampling design, and model assumptions. To make such judgements more reliable, researchers typically follow a workflow—from posing a question and formulating hypotheses and predictions, to designing a study, collecting data, fitting statistical models, and interpreting and communicating results. Yet repeated methodological critiques across disciplines have highlighted persistent weaknesses in this scientific workflow, including vague or weakly testable hypotheses, insufficient sample sizes, and an overreliance on null-hypothesis significance testing that promotes dichotomous “significant / non-significant” interpretations and selective reporting.

Here we propose a small but practical addition to the workflow: a “simulation loop” implemented before data collection (or early in analysis) to make assumptions explicit and to anticipate what the planned data can and cannot reveal. The loop has three steps: (1) translate a conceptual hypothesis into an explicit generative (mechanistic) model; (2) simulate and visualise data implied by that model to scrutinise assumptions about effect sizes, variability, and measurement limitations; and (3) iterate design choices (e.g. sample size, treatment levels, the range of predictor values, measurement error, model structure) using criteria focused on estimation quality—especially interval widths (precision), bias, and robustness to plausible misspecification—rather than binary detect/not-detect framing.

We provide worked examples with reproducible R code, including linear and non-linear cases, and show how the simulation loop can reveal identifiability limits, guide sampling toward decision-relevant precision, and increase transparency in modelling choices.

We argue that these benefits become increasingly important as ecological analyses draw on larger datasets and AI-assisted pattern discovery, where deciding what to believe depends on clearly stated assumptions and defensible uncertainty quantification.

Keywords: Bayesian statistics, statistical methods, quantitative ecology, hypothesis testing, scientific reasoning, experimental design, power analysis

1 Introduction

Many of us have seen the same scene in a conference talk, an in-house seminar, or even in papers. A speaker or author presents a compelling story, shows a clean figure with the mean response in a control and a treatment group (with error bars), and then points to the familiar stars of significance, concluding that the treatment “has an effect” ($p < 0.05$)—or, more problematically, that it “has no effect” ($p > 0.05$). The combination of a coherent narrative, an intuitive plot, and a thresholded summary can make the conclusion feel settled, even though the same data are often compatible with a wide range of effect sizes and uncertainties. For many listeners (and readers), this leaves a quieter set of questions: how much do we actually know from these data, what else are they compatible with, and would the interpretation change under a different but still reasonable analysis?

By default, many of us still use—and are often still taught to use—a binary, threshold-based style of statistical reasoning that originated about a century ago (Gigerenzer, 2004; Neyman & Pearson, 1933) and has been criticised for much of that time, with especially clear calls in recent decades to move beyond “statistical significance” as a scientific standard (e.g. (Anderson *et al.*, 2000; Amrhein *et al.*, 2017; Wasserstein *et al.*, 2019; Amrhein *et al.*, 2019)). Yet in many fields, including ecology and evolutionary biology, practical alternatives that are widely taught, routinely applied, and consistently rewarded remain unevenly implemented. One reason is that compelling stories and crisp conclusions are cognitively and socially attractive, especially when results must be communicated quickly. Another reason is that human evolution has equipped us with a hypersensitive pattern detector and a suite of cognitive biases—such as confirmation bias and motivated reasoning—that can make us overinterpret noisy evidence (Tversky & Kahneman 1974; Nickerson 1998; Kunda 1990; see also error management theory Haselton & Buss 2000). Finally, ecology often provides weaker corrective feedback than fields such as medical science: causal mechanisms are hard to isolate, replication is uncommon, and the consequences of overconfident conclusions are typically slower, more diffuse, and less directly penalised than in clinical contexts where decisions can have immediate, high-stakes outcomes (e.g. see Singal *et al.* 2017; Gelman *et al.* 2019).

Here we propose a small but practical addition to the scientific workflow: a “simulation loop” that is implemented before data collection (or early in analysis) to make assumptions explicit and to explore what the planned data can and cannot reveal. The loop consists of three steps: (1) translate a conceptual hypothesis into an explicit generative (mechanistic) model; (2) simulate and visualise data implied by that model to examine assumptions about effect sizes, variability, and measurement limitations; and (3) iterate design choices (e.g. sample size, treatment levels, the range of predictor values, and model structure) using criteria focused on estimation quality—especially interval widths (precision), bias, and sensitivity to realistic departures from the assumed model—rather than binary detect/not-detect framing. Simulation itself is not new in ecology and evolutionary biology, but it is still used unevenly and is often treated as a specialised add-on rather than a routine step in study design and scientific reasoning. Our contribution is to formalise a simple, teachable workflow that integrates simulation early and makes it easier to apply in everyday ecological research. Throughout the paper we provide worked examples with reproducible R code, including linear and non-linear cases, to show how the simulation loop clarifies assumptions, exposes limits of what the data can support, and supports planning for useful precision. We hope this encourages researchers and students to adopt simulation more routinely as a practical tool for doing better science.

To provide context, Figure 1 summarises common problems and pitfalls that can arise along the traditional scientific workflow. The main pitfalls are briefly explained in box 1, along with key references. None of these issues are new; they have been discussed extensively across statistics and many applied fields, and there is broad agreement that robust scientific conclusions require more than threshold-based summaries or default analysis routines. For brevity, we summarise the main pitfalls in Box 1, which provides brief definitions and key references.

93 2 The simulation loop

94 Here we propose a small but practical addition to the traditional scientific workflow (Figure 1): a
 95 simulation loop that is implemented before data collection (or early in analysis). The aim is to make
 96 assumptions explicit, to translate verbal hypotheses into quantitative expectations, and to explore
 97 what the planned data and model can support under realistic variability and measurement limitations.

99 2.1 Core idea: simulate from an explicit data-generating model

100 At the centre of the simulation loop is an explicit data-generating model. A useful starting point is to
 101 separate the systematic component (pattern) from the stochastic component (noise):

$$y_i = f(x_i; \theta) + \varepsilon_i \quad (1)$$

102 where $f(x_i; \theta)$ is the expected value of y_i given predictors x_i and parameters θ (the signal implied by
 103 the hypothesis), and ε_i is the remaining deviation around that expectation (variation not captured
 104 by the model). Depending on context, ε_i may represent process variation, observation/measurement
 105 error, or both. Importantly, ‘noise’ is not inherently meaningless; it is whatever remains after account-
 106 ing for the structure we chose to model.

107
 108 In addition to specifying how y is generated conditional on x , it is often useful to specify how x
 109 arises under the planned design (e.g. treatment allocation, predictor range, spatial layout, measure-
 110 ment schedule). Making both parts explicit clarifies which features of the data are determined by
 111 design choices and which are left to chance. By simulating from the full data-generating process, we
 112 can explore what patterns we should expect to see if our hypothesis and assumptions were approxi-
 113 mately correct—and what range of outcomes could plausibly arise from variability and measurement
 114 limitations alone.

115 2.2 The three steps of the simulation loop

116 Step 1: From a verbal hypothesis to a mechanistic (mathematical) model.

117 Models are hypotheses. With each model we apply to our data, we examine—often without full
 118 awareness—an additional hypothesis. To constrain and justify this process to a reasonable set of
 119 hypotheses, we propose to build mechanistic models by incorporating existing knowledge about the in-
 120 fluencing variables and their relationships in question. Which explanatory variables are influencing y ?
 121 Is the relationship linear or non-linear? What do we know from the literature? This step nudges us to
 122 think carefully about our hypothesis in terms of model parameters, translating subjective and concep-
 123 tual hypotheses into explicit mathematical formulations. In our experience, a conceptual hypothesis
 124 often intuitively makes sense until we try to translate it into a mechanistic model; then it can become
 125 clear that the original hypothesis lacks direction, concreteness, or foundation in basic knowledge. Be-
 126 yond the classical descriptive hypothesis, a mechanistic model also forces us to think about what we
 127 actually want to estimate, and what values we expect for those parameters (e.g. effect sizes and their
 128 variance). This gives us the opportunity to state hypotheses with numbers and formulas. In ecology,
 129 this often includes anticipating common design features such as hierarchy and non-independence (e.g.
 130 repeated measurements, plots nested within sites), and representing them explicitly in the model.

132 Step 2: Simulate and visualise synthetic (‘fake’) data.

133 “The greatest value of a picture is when it forces us to notice what we never expected to see.” This
 134 quote from Tukey (1977) captures the motivation for the second step. With the mechanistic model

sorted out, we generate a synthetic dataset from it. The key point is not to simulate only a marginal distribution for y , but to simulate y through the model structure: we generate predictor values under the planned design, compute the expected response via $f(x; \theta)$, and then add stochasticity via ε to represent variability and measurement limitations. By plotting the generated data, we see our hypothesis in a concrete, visualised form—not only the direction of an effect, but also its magnitude (e.g. effect size) and consistency (variance). We believe that many ecological papers would benefit from visualising mechanistic model predictions to validate and communicate hypotheses more effectively.

Step 3: Model exploration and design iteration.

The third step is to explore what the model and planned design typically produce when we repeat the study many times on the computer. This goes beyond a classical power analysis that targets a significance threshold. Instead, we repeatedly simulate data under plausible assumptions, fit the planned statistical model, and examine parameter estimates and uncertainty. Does the model behave as we think it does? Do the simulated data show realistic variation? How do estimates and uncertainty intervals change with sample size, predictor range, measurement error, or alternative model structures? We can also fit the model to the simulated data and check whether parameter estimates recover the values we used to generate the data. Going back and forth between assumptions, simulation, and fitting is a playful but powerful way to design a study, understand limitations, and avoid overconfident interpretation before analysing empirical data.

In the following sections we provide concrete examples and reproducible R code to illustrate how this procedure can be applied in practice.

3 Practical example (linear): Plant growth and nitrogen

Data simulation is best shown in practice. The example presented here is deliberately simple and can be reproduced directly. A fully commented R script containing all code shown below (including the simulation loops and figures) can be downloaded from the Supplement.

What influences y in what way?

Every study starts with a research question. Here: What is the influence of nitrogen fertilization on plant growth? We might perform an experiment on sunflowers and fertilize some with nitrogen to assess biomass after one season. Our response variable y is biomass, and a verbal hypothesis could be that plants grow more when nitrogen increases.

To translate this hypothesis into a mechanistic (mathematical) model, we assume a positive linear relationship between biomass and nitrogen concentration x . Here, the “mechanistic” component is intentionally simple and phenomenological; later examples extend the same workflow to explicitly non-linear response functions. A simple model is:

$$y_i = a + bx_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma) \quad (2)$$

where a is the intercept (expected biomass at $x = 0$ on the chosen scale), b is the slope (expected biomass increase per unit nitrogen; an effect size), and σ captures variability around the mean relationship (process variation and/or measurement error).

Setting model parameters

This is where things get interesting and concrete: What do we think these model parameters should look like? In other words, what biomass do we expect at low nitrogen, how strong is the nitrogen effect, and how variable is the relationship?

For illustration, we choose:

```

178     # Specify model parameters
179     a <- 30      # intercept
180     b <- 7       # slope (effect size)
181     sigma <- 5   # SD of residual variation
182

```

These values are not “truth”; they make our assumptions explicit. Later, we can explore how conclusions change when we vary them.

Simulate!

To simulate, we specify a sample size and generate a plausible range of nitrogen concentrations. Here we treat nitrogen as a continuous gradient:

```

189     # Choose a design and simulate data
190     n <- 50
191     x <- rnorm(n, mean = 10, sd = 4)      # nitrogen concentrations
192     y <- a + b*x + rnorm(n, mean = 0, sd = sigma)
193
194
195     fake <- data.frame(x = x, y = y)
196

```

In experiments with fixed treatment levels (rather than a gradient), x would be set to those levels with replication per level, rather than drawn from a distribution. The same simulation logic applies; only the design for generating x changes.

Visualize and assess your fake data

Now we plot the generated data and check whether it matches what we had in mind. This is the hypothesis in visual form, including both effect size and variability.

```

203     plot(fake$x, fake$y,
204           xlab = "Nitrogen concentration (x)",
205           ylab = "Biomass (y)",
206           main = "Simulated (fake) data")
207

```

We can also fit the planned model and check whether the estimated parameters are in the range we assumed. This is a quick sanity check that the analysis can recover the generating values under these assumptions.

```

212     fit_lm <- lm(y ~ x, data = fake)
213     summary(fit_lm)
214     confint(fit_lm) # 95% confidence intervals
215
216
217     plot(fake$x, fake$y,
218           xlab = "Nitrogen concentration (x)",
219           ylab = "Biomass (y)",
220           main = "Simulated data with fitted regression line")
221     abline(fit_lm)
222

```

At this point, it is tempting to focus on a p-value. In the simulation loop, we instead look at the estimates and their uncertainty—for example, the width of the 95% confidence interval for b .

How to use this — Play!

Now we go back and forth and explore what changes when we manipulate assumptions and design choices. A simple and practical target is precision: how wide are uncertainty intervals for the parameter

we care about (here the nitrogen effect b), and how does that depend on sample size, the range of nitrogen values, or the amount of variability and measurement error?

The code below repeats the simulated study many times and summarises the width of the 95% confidence interval for b across repetitions. These design-precision curves are conditional on the assumed model (linearity, error distribution, and constant variance); later examples show how to relax these assumptions and test sensitivity.

```
# One simulation run: generate data, fit model, return CI width for slope
# b
one_run <- function(n, a, b, sigma, mean_x = 10, sd_x = 4) {
  x <- rnorm(n, mean = mean_x, sd = sd_x)
  y <- a + b*x + rnorm(n, 0, sigma)
  fit <- lm(y ~ x)
  ci <- confint(fit)["x", ]
  return(ci[2] - ci[1])
}

set.seed(1)

# Precision vs sample size
ns <- c(10, 20, 30, 50, 80, 120)
R <- 500

out_n <- data.frame(n = ns, mean_width = NA, sd_width = NA)
for (j in seq_along(ns)) {
  widths <- replicate(R, one_run(ns[j], a, b, sigma))
  out_n$mean_width[j] <- mean(widths)
  out_n$sd_width[j] <- sd(widths)
}

plot(out_n$n, out_n$mean_width, type = "b",
     xlab = "Sample size (n)",
     ylab = "Mean 95% CI width for slope (b)",
     main = "Precision of the nitrogen effect vs sample size")
```

If a study requires the nitrogen effect to be estimated within $\pm k$ biomass units per nitrogen unit, then a rough design target is a 95% confidence interval width of about $2k$ for b . The simulation loop provides a direct way to choose n (and other design choices) to meet such a target under stated assumptions. A second lever in regression is the range of the predictor: even with the same n , a narrow nitrogen range makes the slope harder to estimate. We can explore that directly:

```
# Precision vs predictor range (sd of x), at fixed n
sd_x_values <- c(1, 2, 4, 6)
n_fixed <- 50

out_x <- data.frame(sd_x = sd_x_values, mean_width = NA)
for (k in seq_along(sd_x_values)) {
  widths <- replicate(R, one_run(n_fixed, a, b, sigma,
    mean_x = 10, sd_x = sd_x_values[k]))
  out_x$mean_width[k] <- mean(widths)
}

plot(out_x$sd_x, out_x$mean_width, type = "b",
     xlab = "SD of nitrogen values (predictor range)",
     ylab = "Mean 95% CI width for slope (b)",
     main = "Precision depends on predictor range")
```

Finally, we can explore how increased variability and measurement error affect precision. For example, if the residual standard deviation doubles, uncertainty intervals widen substantially under the same design:

```
# Sensitivity to variability / measurement error: larger sigma
sigma2 <- 10

out_n2 <- data.frame(n = ns, mean_width = NA)
for (j in seq_along(ns)) {
  widths <- replicate(R, one_run(ns[j], a, b, sigma2))
  out_n2$mean_width[j] <- mean(widths)
}

plot(out_n2$n, out_n2$mean_width, type = "b",
     xlab = "Sample size (n)",
     ylab = "Mean 95% CI width for slope (b)",
     main = "Precision vs sample size under different variability")
lines(out_n2$n, out_n2$mean_width, type = "b", lty = 2)
legend("topright", legend = c(paste0("sigma = ", sigma),
                             paste0("sigma = ", sigma2)),
     lty = c(1, 2), bty = "n")
```

This back-and-forth is the core of the simulation loop in practice: we make assumptions explicit, generate the data those assumptions imply, and then explore how design choices translate into the uncertainty of our key estimates. Improving precision, however, does not by itself make an association causal; the loop complements, but does not replace, careful causal reasoning and design. In the next section, we extend the same workflow to non-linear relationships, where model structure and identifiability become even more central.

4 Practical example (non-linear): Saturating growth response to nitrogen

Many ecological relationships are non-linear: responses can saturate, show thresholds, or peak at intermediate conditions. In such cases, simulation is especially useful because the same data can often be explained by different parameter combinations, and identifiability can depend strongly on the predictor range and the design. Below we extend the same workflow to a simple saturating response of biomass to nitrogen, using a Michaelis–Menten (rectangular hyperbola) function.

What influences y in what way?

We keep the same research question—how nitrogen influences plant growth—but now assume that growth increases with nitrogen and then levels off as other resources become limiting. A simple non-linear model is:

$$y_i = \frac{V_{\max}x_i}{K + x_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma) \quad (3)$$

where $V_{\max}$ is the asymptotic biomass (maximum expected response as $x \rightarrow \infty$), K is the half-saturation constant (the nitrogen value at which expected biomass is $V_{\max}/2$), and σ captures variability and measurement error around the mean response.

Setting model parameters

As before, we must state plausible parameter values. For illustration, we choose:

```

329     # Non-linear model parameters (Michaelis-Menten)
330     Vmax <- 120    # asymptotic biomass
331     K <- 8        # half-saturation constant
332     sigma <- 10   # residual SD
333

```

335 Simulate!

336 A key design choice in non-linear models is the range of x . To show why this matters, we simulate two
 337 designs with the same sample size: one that samples nitrogen only in a low-to-moderate range (where
 338 saturation is not well observed), and one that samples a wider range that includes values high enough
 339 to approach the asymptote.

```

340     # Helper: Michaelis-Menten mean function
341     mu_mm <- function(x, Vmax, K) Vmax * x / (K + x)
342
343
344     set.seed(2)
345     n <- 50
346
347     # Design A: narrow nitrogen range (poor leverage on asymptote)
348     x_narrow <- runif(n, min = 0, max = 12)
349
350     # Design B: wide nitrogen range (better leverage on asymptote)
351     x_wide <- runif(n, min = 0, max = 40)
352
353     # Simulate data
354     y_narrow <- mu_mm(x_narrow, Vmax, K) + rnorm(n, 0, sigma)
355     y_wide <- mu_mm(x_wide, Vmax, K) + rnorm(n, 0, sigma)
356
357     fake_narrow <- data.frame(x = x_narrow, y = y_narrow)
358     fake_wide <- data.frame(x = x_wide, y = y_wide)
359

```

360 Visualize and assess your fake data

361 Plotting the two simulated datasets immediately shows why the design matters: in the narrow range,
 362 the relationship can look almost linear, whereas the wide range reveals curvature and the onset of
 363 saturation.

```

364     plot(fake_narrow$x, fake_narrow$y,
365          xlab = "Nitrogen (x)", ylab = "Biomass (y)",
366          main = "Non-linear model: narrow nitrogen range")
367     curve(mu_mm(x, Vmax, K), add = TRUE)
368
369     plot(fake_wide$x, fake_wide$y,
370          xlab = "Nitrogen (x)", ylab = "Biomass (y)",
371          main = "Non-linear model: wide nitrogen range")
372     curve(mu_mm(x, Vmax, K), add = TRUE)
373

```

375 How to use this — Play!

376 We now fit the planned non-linear model and examine what it typically produces under each design.
 377 In base R, a simple option is `nls` (non-linear least squares). Non-linear fitting can be sensitive to
 378 starting values, so we provide reasonable starts.


```

379   # Fit non-linear model with nls
380   fit_mm <- function(dat) {
381     nls(y ~ Vmax * x / (K + x),
382       data = dat,
383       start = list(Vmax = max(dat$y), K = median(dat$x)),
384       control = nls.control(maxiter = 200, warnOnly = TRUE))
385   }
386
387   fitA <- fit_mm(fake_narrow)
388   fitB <- fit_mm(fake_wide)
389
390   summary(fitA)
391   summary(fitB)
392
393

```

394 Even when `nls` converges, the key question is not whether parameters exist, but how uncertain they
395 are and how much they depend on design choices. A simple, comparable metric is again interval width.
396 For `nls`, we can approximate confidence intervals for parameters. (In some datasets, `confint` may fail
397 or be slow; in that case, repeating the simulation and examining the spread of estimates provides the
398 same message.)

```

399   # Approximate 95% confidence intervals for parameters (may be slow for
400   some fits)
401   confint(fitA)
402   confint(fitB)
403
404

```

405 To make the comparison more robust and aligned with the simulation loop, we repeat the whole
406 study many times under each design and summarise the distribution of parameter estimates. This
407 makes identifiability issues visible: if many different ($V_{\max}$, K) combinations fit the data similarly,
408 the estimates will vary widely across repetitions, especially under the narrow-range design.

```

409   one_run_mm <- function(n, x_max, Vmax, K, sigma) {
410     x <- runif(n, 0, x_max)
411     y <- mu_mm(x, Vmax, K) + rnorm(n, 0, sigma)
412     dat <- data.frame(x = x, y = y)
413
414     fit <- try(fit_mm(dat), silent = TRUE)
415     if (inherits(fit, "try-error")) return(c(Vmax_hat = NA, K_hat = NA))
416
417     co <- coef(fit)
418     c(Vmax_hat = unname(co["Vmax"]), K_hat = unname(co["K"]))
419   }
420
421   set.seed(3)
422   R <- 300
423   n <- 50
424
425   # Design A: x_max = 12 (narrow); Design B: x_max = 40 (wide)
426   estA <- replicate(R, one_run_mm(n, x_max = 12, Vmax, K, sigma))
427   estB <- replicate(R, one_run_mm(n, x_max = 40, Vmax, K, sigma))
428
429   estA <- t(estA); estB <- t(estB)
430   estA <- as.data.frame(estA); estB <- as.data.frame(estB)
431
432   # Remove failed fits
433   estA <- estA[complete.cases(estA), ]
434
435

```

```

436     estB <- estB[complete.cases(estB), ]
437
438     # Compare distributions of estimates
439     summary(estA)
440     summary(estB)
441
442     plot(estA$K_hat, estA$Vmax_hat,
443          xlab = "K estimate", ylab = "Vmax estimate",
444          main = "Narrow range: strong parameter trade-off")
445     abline(v = K, h = Vmax, lty = 2)
446
447     plot(estB$K_hat, estB$Vmax_hat,
448          xlab = "K estimate", ylab = "Vmax estimate",
449          main = "Wide range: improved identifiability")
450     abline(v = K, h = Vmax, lty = 2)
451

```

452 This “cloud of estimates” is a simple diagnostic: in the narrow-range design, $V_{\max}$ and K often trade
453 off against each other (many combinations produce similar curves over $x \in [0, 12]$), so parameter
454 uncertainty can remain large even with the same sample size. In contrast, including higher nitrogen
455 values provides leverage to distinguish curvature and the asymptote, improving identifiability and
456 reducing uncertainty.

457 Finally, we can connect this directly back to design planning by comparing a precision proxy (e.g. the
458 interquartile range of estimates or the spread of fitted parameters) across choices of n and x -range.
459 For example:

```

460     # Quick design exploration: how does n affect the spread of K estimates?
461     ns <- c(20, 50, 100)
462     R <- 200
463
464
465     spread <- function(n, x_max) {
466         est <- replicate(R, one_run_mm(n, x_max, Vmax, K, sigma))
467         est <- t(est)
468         est <- est[complete.cases(est), , drop = FALSE]
469         # use IQR as a robust spread measure
470         c(IQR_Vmax = IQR(est[, "Vmax_hat"]), IQR_K = IQR(est[, "K_hat"]))
471     }
472
473     out <- expand.grid(n = ns, x_max = c(12, 40))
474     out$IQR_Vmax <- NA
475     out$IQR_K <- NA
476
477     for (i in seq_len(nrow(out))) {
478         s <- spread(out$n[i], out$x_max[i])
479         out$IQR_Vmax[i] <- s["IQR_Vmax"]
480         out$IQR_K[i] <- s["IQR_K"]
481     }
482
483     out
484
485     plot(out$n[out$x_max == 12], out$IQR_K[out$x_max == 12], type = "b",
486          xlab = "Sample size (n)", ylab = "IQR of K estimates",
487          main = "Precision depends on n and x-range")
488     lines(out$n[out$x_max == 40], out$IQR_K[out$x_max == 40], type = "b", lty
489           = 2)
490     legend("topright", legend = c("x in [0,12]", "x in [0,40]"),
491           lty = c(1,2), bty = "n")
492

```

493 This non-linear example illustrates why simulation is especially valuable beyond linear settings: identi-
494 fiability and precision can depend as much on *where* you sample (predictor range and design) as on how
495 many observations you collect. In the next section, we generalise these ideas and briefly discuss how
496 simulation-based thinking becomes even more important when models grow more complex (e.g. mul-
497 tiple predictors, interactions, hierarchical designs) and when inference relies on additional modelling
498 assumptions.

499 5 Practical guidance: implementing the simulation loop

500 This section summarises practical guidance to implement the simulation loop in everyday ecological
501 research. The goal is not to add another layer of complexity, but to make assumptions explicit, align
502 study design with the parameters we care about, and report simulation results transparently and
503 reproducibly.

504 5.1 Practical guidance (short)

505 **Choose a generative model that matches the question.** Start from the verbal hypothesis and
506 translate it into a simple generative model: define y , the predictors/design variables x , the expected
507 response $f(x; \theta)$, and the stochastic component ε (process variation and/or measurement error). Begin
508 simple and add complexity only when it changes the inference you want to make (e.g. hierarchy,
509 autocorrelation, heterogeneity, detection).

510 **State plausible parameter values (and ranges).** Simulation is only as informative as the assump-
511 tions it explores. Use prior knowledge, the literature, pilot data, or clearly stated expert judgement
512 to propose plausible values for effect sizes, variance components, and measurement limitations. When
513 unsure, explore a range of values and report how conclusions depend on them.

514 **Simulate from the design you actually plan to run.** Explicitly represent the sampling design
515 (treatment levels, replication structure, predictor range, blocking, repeated measures). In non-linear
516 settings, pay special attention to the predictor range: identifiability can depend as much on *where* you
517 sample as on how many observations you collect.

518 **Focus Step 3 on estimation, not thresholds.** Treat Step 3 as “playing through” the planned study
519 many times on the computer. Fit the planned analysis model and summarise what you typically get in
520 terms of parameter estimates and uncertainty—especially uncertainty-interval widths (precision). Use
521 bias and sensitivity to realistic departures from assumptions (e.g. increased noise, mild misspecification,
522 measurement error) as additional diagnostics. This does not guarantee correct inference; it makes
523 explicit what follows from the assumptions you are willing to make, and how sensitive results are to
524 them.

525 **Decide design targets in terms of precision and relevance.** If the scientific or management
526 claim requires an effect to be estimated within $\pm k$ units on the chosen scale, plan for an interval width
527 of about $2k$ for the relevant parameter. Use simulation outputs (design-precision curves) to choose n ,
528 predictor range, and measurement effort that meet this target under plausible assumptions.

529 **Report what you simulated and why.** To make the workflow useful (and teachable), report the
530 generative model, key assumptions, the range of parameters explored, the performance criterion (e.g.
531 interval width), and the code required to reproduce results.

Checklist: reportable outputs of the simulation loop

- **Verbal hypothesis** → **model**: equations for $f(x; \theta)$ and the stochastic component (ε), including the response scale.
- **Assumptions**: plausible parameter values (and ranges) with justification (literature, pilot data, expert judgement).
- **Stochastic structure**: what ε represents (process variation, measurement error, both), and whether hierarchy/autocorrelation is included.
- **Design explored**: n , treatment levels, predictor range, replication structure, measurement effort; what was varied and why.
- **Performance target**: interval width (precision) for key parameters; optionally bias and sensitivity to departures from assumptions.
- **Summary outputs**: design-precision curve(s) (e.g. interval width vs n , vs predictor range, vs noise).
- **Reproducibility**: code (and seed) to reproduce simulations and figures (e.g. in Supplement), plus software versions if relevant.

532

5.2 Common extensions (brief)

533

534 The simulation loop is not restricted to linear regression. Common extensions follow the same three
535 steps but use a different generative model and analysis model:

536 **Two-group designs.** A two-level treatment factor is a special case: simulate group means and
537 variability, then examine the precision of the treatment contrast (difference in means) as a function of
538 sample size and measurement error.

539 **Generalised responses.** For counts, proportions, presence/absence, and rates, specify an appropriate
540 observation model (e.g. Poisson/negative binomial, binomial) and simulate on the natural scale.
541 Precision targets may be set on the link scale or transformed back to the response scale.

542 **Hierarchical and repeated-measures designs.** Simulate nested structure (e.g. plots within sites,
543 repeated measurements through time) and fit models with matching random effects or correlation
544 structure. Examine how effective sample size and precision depend on the number of groups and
545 replicates per group.

546 **Non-linear and mechanistic models.** When responses saturate, show thresholds, or depend on inter-
547 acting processes, simulate under plausible parameter ranges and ensure the design includes predictor
548 values that make key parameters identifiable (as illustrated in the non-linear example above).

549 **Model checking after fitting.** Simulation can also be used after model fitting (e.g. posterior pre-
550 dictive checks). In this paper we emphasise simulation earlier, when it can still inform study design
551 and the formulation of quantitative hypotheses.

6 Outlook: why the simulation loop matters (especially with AI)

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553

554 This paper offers a compact way to make simulation part of routine ecological inference: turn a verbal
555 hypothesis into an explicit generative model, simulate what it implies, and iterate design and modelling
556 choices based on the estimates you would obtain and their uncertainty, rather than on whether a test
557 crosses an arbitrary threshold.

558 A practical motivation is that different analysis choices sometimes lead to similar conclusions, but
559 sometimes they do not. When results are fragile to reasonable alternatives (e.g. link functions, dis-

tributional assumptions, treatment of non-independence, or measurement error), it becomes hard to know what to believe. The simulation loop provides a constructive way to examine this: simulate under plausible data-generating processes, fit candidate analyses, and quantify how estimates and uncertainty change across modelling choices. In that sense, simulation is not only a design tool but also a diagnostic for robustness and identifiability (see also simulation-based approaches to understanding and validating complex models in ecology; DiRenzo *et al.* 2023).

More broadly, choosing an analysis and deciding what counts as evidence is unavoidably epistemic: it depends on background assumptions about how data are generated, what uncertainty means, and what kinds of claims are warranted in the face of error and limited information. These questions are actively studied in philosophy of science and the foundations of statistics, but they often remain implicit in everyday practice (Romeijn, 2014; Gelman & Shalizi, 2013). Making the generative model explicit turns many hidden choices into inspectable assumptions. It also clarifies that (i) precise estimates can still be wrong when key mechanisms or error processes are misspecified (Box, 1976), and (ii) methodological choices can legitimately reflect “inductive risk”—the consequences of different kinds of error—which motivates being explicit about what uncertainty is acceptable for a given claim (Douglas, 2000; Steel, 2010). In ecology and evolution, this perspective aligns with growing calls to improve uncertainty consideration and propagation beyond narrow reliance on single metrics, and to adopt clearer standards for reporting uncertainty and the consequences of model structure (Simmonds *et al.*, 2024; Lele, 2020).

These benefits become increasingly important as ecology and evolution incorporate larger datasets, more complex models, and AI-assisted pattern discovery. Modern tools can detect subtle regularities in high-dimensional data, but they do not remove the need to decide what to believe and what uncertainty is defensible. Simulation-based thinking provides a practical way to expose and communicate assumptions, to quantify the consequences of modelling choices, and to distinguish predictive success from mechanistic understanding. In this sense, the simulation loop is both a design tool and a reasoning tool: it helps researchers and students move from compelling stories toward quantitative expectations, transparent uncertainty, and conclusions that are proportional to what the data can support.

At the same time, simulation does not guarantee correct inference: its value is conditional on the assumptions explored. It cannot by itself resolve confounding in observational studies or substitute for careful causal design (Pearl, 2019). But it can make these limitations explicit—for example, by showing that even very large datasets can yield narrow uncertainty intervals for the *wrong* quantity if key mechanisms or measurement processes are misspecified. This matters especially in AI-enabled workflows, where model complexity and opacity can make it difficult to understand how conclusions were produced, and where standards for uncertainty quantification and error propagation remain uneven across tools and subfields (Pichler & Hartig, 2023; Lucas, 2020). By encouraging explicit generative assumptions and transparent reporting of their consequences, the simulation loop offers a practical step toward more defensible inference in a future where pattern discovery will be easy, but deciding what to believe may be harder than ever.

References

- Amrhein, V., Gelman, A., Greenland, S. & McShane, B.B. (2019). Abandoning statistical significance is both sensible and practical. Tech. Rep. e27657v1, PeerJ Inc.
- Amrhein, V., Korner-Nievergelt, F. & Roth, T. (2017). The earth is flat ($p > 0.05$): Significance thresholds and the crisis of unreplicable research. *PeerJ*, 5, e3544.
- Anderson, D.R., Burnham, K.P. & Thompson, W.L. (2000). Null Hypothesis Testing: Problems, Prevalence, and an Alternative. *The Journal of Wildlife Management*, 64, 912–923.
- Baker, M. (2016). 1,500 scientists lift the lid on reproducibility. *Nature*, 533, 452–454.
- Barnett, A.G., van der Pols, J.C. & Dobson, A.J. (2005). Regression to the mean: What it is and how to deal with it. *International Journal of Epidemiology*, 34, 215–220.
- Box, G.E.P. (1976). Science and Statistics. *Journal of the American Statistical Association*, 71, 791–799.
- Camerer, C.F., Dreber, A., Holzmeister, F., Ho, T.H., Huber, J., Johannesson, M., Kirchler, M., Nave, G., Nosek, B.A., Pfeiffer, T., Altmejd, A., Buttrick, N., Chan, T., Chen, Y., Forsell, E., Gampa, A., Heikensten, E., Hummer, L., Imai, T., Isaksson, S., Manfredi, D., Rose, J., Wagenmakers, E.J. & Wu, H. (2018). Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2, 637–644.
- Cumming, G. (2013). *Understanding the New Statistics: Effect Sizes, Confidence Intervals, and Meta-Analysis*. Routledge.
- DiRenzo, G.V., Hanks, E. & Miller, D.A.W. (2023). A practical guide to understanding and validating complex models using data simulations. *Methods in Ecology and Evolution*, 14, 203–217.
- Douglas, H. (2000). Inductive Risk and Values in Science. *Philosophy of Science*, 67, 559–579.
- Gelman, A., Carlin, J.B. & Nallamothu, B.K. (2019). ORBITA and coronary stents: A case study in the analysis and reporting of clinical trials.
- Gelman, A. & Loken, E. (2013). The garden of forking paths: Why multiple comparisons can be a problem, even when there is no “fishing expedition” or “p-hacking” and the research hypothesis was posited ahead of time. *Department of Statistics, Columbia University*, 348, 3.
- Gelman, A. & Shalizi, C.R. (2013). Philosophy and the practice of Bayesian statistics. *British Journal of Mathematical and Statistical Psychology*, 66, 8–38.
- Gigerenzer, G. (2004). Mindless statistics. *The Journal of Socio-Economics*, 33, 587–606.
- Goodman, S.N. & Berlin, J.A. (1994). The Use of Predicted Confidence Intervals When Planning Experiments and the Misuse of Power When Interpreting Results. *Annals of Internal Medicine*, 121, 200–206.
- Halsey, L.G., Curran-Everett, D., Vowler, S.L. & Drummond, G.B. (2015). The fickle P value generates irreproducible results. *Nature Methods*, 12, 179–185.
- Haselton, M.G. & Buss, D.M. (2000). Error management theory: A new perspective on biases in cross-sex mind reading. *Journal of Personality and Social Psychology*, 78, 81–91.
- Ioannidis, J.P.A. (2005). Why Most Published Research Findings Are False. *PLOS Medicine*, 2, e124.
- Kerr, N.L. (1998). HARKing: Hypothesizing After the Results are Known. *Personality and Social Psychology Review*, 2, 196–217.

- Kunda, Z. (1990). The case for motivated reasoning. *Psychological Bulletin*, 108, 480–498.
- Lele, S.R. (2020). How Should We Quantify Uncertainty in Statistical Inference? *Frontiers in Ecology and Evolution*, 8.
- Lucas, T.C.D. (2020). A translucent box: Interpretable machine learning in ecology. *Ecological Monographs*, 90, e01422.
- Muff, S., Nilsen, E.B., O’Hara, R.B. & Nater, C.R. (2022). Rewriting results sections in the language of evidence. *Trends in Ecology & Evolution*, 37, 203–210.
- Neyman, J. & Pearson, E.S. (1933). On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society of London, Series A: Containing Papers of a Mathematical or Physical Character*, 231, 289–337.
- Nickerson, R.S. (1998). Confirmation Bias: A Ubiquitous Phenomenon in Many Guises. *Review of General Psychology*, 2, 175–220.
- Pearl, J. (2019). The seven tools of causal inference, with reflections on machine learning. *Commun. ACM*, 62, 54–60.
- Perezgonzalez, J.D. (2015). Fisher, Neyman-Pearson or NHST? A tutorial for teaching data testing. *Frontiers in Psychology*, 6.
- Pichler, M. & Hartig, F. (2023). Machine learning and deep learning—A review for ecologists. *Methods in Ecology and Evolution*, 14, 994–1016.
- Popovic, G., Mason, T.J., Drobniak, S.M., Marques, T.A., Potts, J., Joo, R., Altwegg, R., Burns, C.C.I., McCarthy, M.A., Johnston, A., Nakagawa, S., McMillan, L., Devarajan, K., Taggart, P.L., Wunderlich, A., Mair, M.M., Martínez-Lanfranco, J.A., Lagisz, M. & Pottier, P. (2024). Four principles for improved statistical ecology. *Methods in Ecology and Evolution*, 15, 266–281.
- Rajtmajer, S.M., Errington, T.M. & Hillary, F.G. (2022). How failure to falsify in high-volume science contributes to the replication crisis. *eLife*, 11, e78830.
- Romeijn, J.W. (2014). The Stanford Encyclopedia of Philosophy.
- Rothman, K.J. & Greenland, S. (2018). Planning Study Size Based on Precision Rather Than Power. *Epidemiology*, 29, 599.
- Simmonds, E.G., Adjei, K.P., Cretois, B., Dickel, L., González-Gil, R., Laverick, J.H., Mandeville, C.P., Mandeville, E.G., Ovaskainen, O., Sicacha-Parada, J., Skarstein, E.S. & O’Hara, B. (2024). Recommendations for quantitative uncertainty consideration in ecology and evolution. *Trends in Ecology & Evolution*, 39, 328–337.
- Singal, D., Chateau, D. & Brownell, M. (2017). Prenatal Antidepressant Use and Autism Spectrum Disorder. *JAMA*, 318, 664–665.
- Steel, D. (2010). Epistemic Values and the Argument from Inductive Risk. *Philosophy of Science*, 77, 14–34.
- Stefan, A.M. & Schönbrodt, F.D. (2023). Big little lies: A compendium and simulation of p-hacking strategies. *Royal Society Open Science*, 10, 220346.
- Tukey, J.W. (1977). Exploratory data analysis. *Reading/Addison-Wesley*.
- Tversky, A. & Kahneman, D. (1974). Judgment under Uncertainty: Heuristics and Biases. *Science*, 185, 1124–1131.

- 679 van Zwet, E.W. & Cator, E.A. (2021). The significance filter, the winner’s curse and the need to
680 shrink. *Statistica Neerlandica*, 75, 437–452.
- 681 Wasserstein, R.L., Schirm, A.L. & Lazar, N.A. (2019). Moving to a World Beyond “ $p < 0.05$ ”. *The*
682 *American Statistician*, 73, 1–19.
- 683 Wolkovich, E.M., Auerbach, J., Chamberlain, C.J., Buonaiuto, D.M., Ettinger, A.K., Morales-Castilla,
684 I. & Gelman, A. (2021). A simple explanation for declining temperature sensitivity with warming.
685 *Global Change Biology*, 27, 4947–4949.
- 686 Yang, Y., Sánchez-Tójar, A., O’Dea, R.E., Noble, D.W.A., Koricheva, J., Jennions, M.D., Parker,
687 T.H., Lagisz, M. & Nakagawa, S. (2023). Publication bias impacts on effect size, statistical power,
688 and magnitude (Type M) and sign (Type S) errors in ecology and evolutionary biology. *BMC*
689 *Biology*, 21, 71.

Box 1: Common pitfalls along the traditional scientific workflow

This box summarises common pitfalls in roughly the order they can arise along the traditional workflow shown in Figure 1. The proposed simulation loop provides a tool to anticipate and prevent many of these pitfalls.

Identifying research question

(1) Confirmation bias and motivated reasoning. Interpreting evidence in ways that align with prior beliefs or preferred narratives. Refs: Nickerson 1998; Kunda 1990; Tversky & Kahneman 1974.

(2) Weakly specified hypotheses and undefined alternatives. Hypotheses may be vague, lack a clear direction or quantitative prediction, and the implied alternative to the null is often left undefined (e.g. “there is a difference”), which makes interpretation and design targets ambiguous. Refs: Rajtmajer *et al.* 2022; Gigerenzer 2004; Perezgonzalez 2015.

(3) Confounding and causal overinterpretation of associations. A statistical association does not, by itself, identify a causal effect. Apparent treatment effects can arise because groups differ in other drivers (confounding), because sampling is selective, or because key variables were not measured. As a result, strong narratives can be attached to patterns that would also be expected under plausible alternative explanations. Refs: Pearl 2019; Rajtmajer *et al.* 2022.

Study design

(4) Insufficient sample sizes. Small samples frequently yield imprecise estimates and unstable results, and can exaggerate effect sizes by chance. Refs: Halsey *et al.* 2015; Ioannidis 2005; Baker 2016; Camerer *et al.* 2018.

(5) Planning for “power to detect” rather than precision/relevance. Sample size decisions that target a significance threshold rather than a decision-relevant interval width or estimation goal. Refs: Goodman & Berlin 1994; Rothman & Greenland 2018; Cumming 2013.

(6) Measurement error, imperfect detection, and misclassification. Noise and bias introduced by measurement instruments, proxies, observer effects, or detection limits can attenuate effects, inflate uncertainty, or create spurious patterns if not anticipated and modeled. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

(7) Non-independence and pseudoreplication. Spatial/temporal autocorrelation, clustering, and repeated measures can reduce the effective sample size and bias uncertainty if treated as independent. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

Analysis

(8) Overreliance on p-values and “statistical significance”. Dichotomous interpretations substitute for effect sizes, uncertainty, and scientific relevance. Refs: Amrhein *et al.* 2017, 2019; Wasserstein *et al.* 2019.

(9) Multiple analysis choices: multiple testing, HARKing, and p-hacking. Many defensible analysis choices and multiple comparisons can yield different outcomes; hypotheses may be adjusted after seeing results (HARKing), and selective exploration can create apparently strong results without deliberate misconduct. Refs: Gelman & Loken 2013; Stefan & Schönbrodt 2023; Kerr 1998.

(10) Mismatch between model and data-generating process. Inference can be distorted when key features of the data are ignored or misrepresented (e.g. non-linear relationships fit as linear, wrong distribution or link, omitted random effects, ignored autocorrelation, inappropriate pooling, or unmodelled measurement error). Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

(11) Mathematical artefacts and scale/transform issues. Patterns can arise from statistical artefacts (e.g. regression to the mean) or inappropriate transformations; these can mimic biological effects. Refs: Barnett *et al.* 2005; Wolkovich *et al.* 2021.

Interpretation & Communication

(12) Overconfident “yes/no” summaries. Binary conclusions can sound decisive even when intervals are wide or model-dependent. Refs: Muff *et al.* 2022; Amrhein *et al.* 2019.

(13) Misreading “non-significant” as “no effect”. Non-significant results are often compatible with effects that are biologically meaningful but imprecisely estimated. Refs: Goodman & Berlin 1994; Rothman & Greenland 2018; Singal *et al.* 2017.

(14) Selective reporting and publication bias (significance filtering). Significant/tidy results are more likely to be emphasized and published, distorting the evidence base. Refs: Yang *et al.* 2023; van Zwet & Cator 2021.

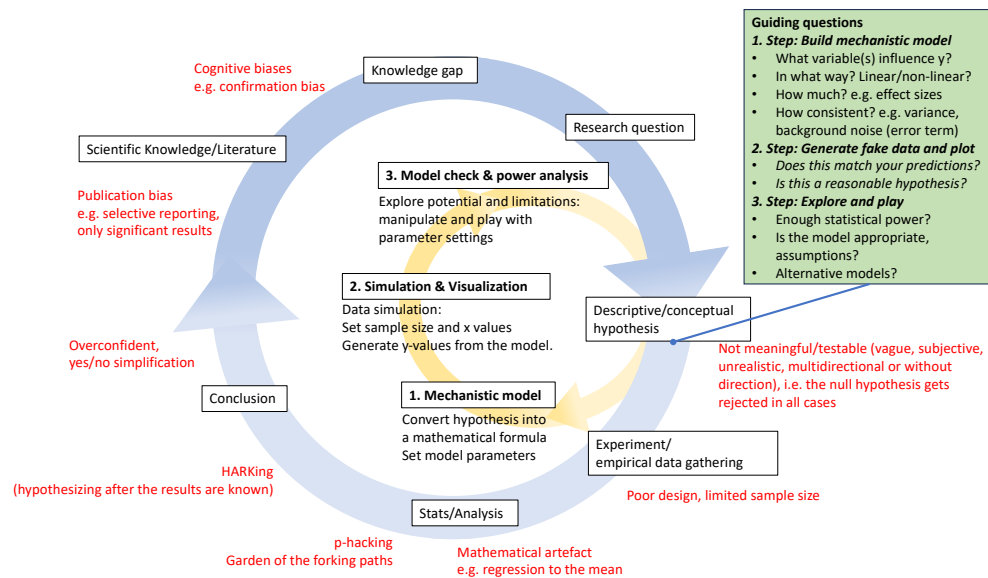


Figure 1: Schematic overview of the current scientific workflow (blue circle) with its related problems (red text) and how some of them can be anticipated with the help of data simulation (three-step-loop in yellow). The critical step is the translation of a verbal hypothesis into a mechanistic model expressed in mathematical terms. We believe that this 'simulation loop' fosters a better understanding of underlying processes founded in theory, clear predictions that can be evaluated and overall better science. The questions summarized in the green box should be addressed in the simulation loop before continuing on the blue circle, e.g. prior to the start of real data gathering/exploration.